CAPSULE RESEARCH PROPOSAL

I. BASIC INFORMATION

Project Title: CATCHFISH: A Collaborative ICT-Based Tools for an Optimize Fishing Repository and Catch Efficiency Towards a Sustainable Use and Development

Study 1: **Implementation of RDBMS in Establishing a Digital Repository for Community Fisheries Information and Regional Knowledge**
Time Frame: January – June 2024

Study 2: **Design and Development of CATCHFISH Front-end Incorporating the Community Participation Modules**
Time Frame: April 2024 – September 2024

Study 3: **Empowering Communities through CATCHFISH Evaluation, Training and Technology Transfer**
Time Frame: October 2024 – December 2024

Proponents:
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Implementing Agency: Partido State University, CAS-Institute of Information Technology
Cooperating Agency: World Wildlife Fund, Philippines
Project Site: Municipalities along Lagonoy Gulf

II. WORK PLAN SUMMARY

Planned Duration of the Study: One-Year
Planned Start Date: January 2024
Planned Completion Date: December 2024

Strategies of Implementation (Gantt Chart):

Objectives	Activities	Output	Year 1			
			Q1	Q2	Q3	Q4
Study 1 - Implementation of RDBMS in Establishing a Digital Repository for Community Fisheries Information and Regional Knowledge						
1. Gathering and compiling data	1. Coordinate with the WWF, BFAR, and other	1. Permission to conduct the study.				

and information about Fish catch and Fisheries communities. 2. Design the conceptual and logical framework for CATCHFISH Repository. 3. Develop the appropriate tables for the CATCHFISH repository covering all the Fish Catch and Fisheries communities' comprehensive resources and information.	key informants of the study 2. Conduct observation, interview, and FGD with the key informant 3. Identify the appropriate tools for developing the CATCHFISH backend. 4. Prepare the conceptual and logical database design. 5. Develop the CATCHFISH backend. 6. Writing of Manuscript.	2. Observation and diary on Fish catch and Fisheries communities. 3. A comprehensive CATCHFISH backend covering logical and physical database components. 4. Draft of publication regarding the comprehensive CATCHFISH backend covering logical and physical database design and components.				
Study 2 - Design and Development of CATCHFISH Front-end Incorporating the Community Participation Modules						
1. Design and develop the CATCHFISH as a collaborative tool for fishing resources and catch efficiency integrating functionalities such as a data entry page for content creation, knowledge retrieval features, social network services, and other viewable elements. 2. Evaluation and unit testing of content creation pages, knowledge retrieval pages, social network services pages, and other	1. Analysis of System Requirements (Hardware, Software, People ware, Policies, etc.) 2. Design and develop the required module or functionalities in CATCHFISH as a collaborative tool for fishing resources and catch efficiency 3. Conduct an evaluation and unit testing for every developed module/functionality. 4. Writing of Manuscript.	1. User Interface Prototype, Process Diagrams, and CRUD functionalities 2. Data retrieval optimizations 3. Functionalities for report generation 4. Online dashboard, management page and monitoring tool. 5. Draft of publication regarding the development life cycle of CATCHFISH.				

viewable elements pages.						
Study 3 - Empowering Communities through CATCHFISH Training and Technology Transfer						
1. Conduct the usability testing of the system using the ISO 25010:2011 quality model. 2. Conduct capability-building activities with the Fisher folks, communities, and other stakeholders. 3. Formal turnover of the developed CATCHFISH to the stakeholder.	1. Conduct a usability evaluation using ISO 25010:2011. 2. Conduct capability building with the target stakeholders 3. Write a manuscript	1. Fisher folks, communities, and other stakeholders are acquainted and familiarized with the system's functionalities, operations, and administration, 2. Complete Package of CATCHFISH				

III. TECHNICAL DESCRIPTION OF THE RESEARCH

Rationale

Food security, as defined by the United Nations' Committee on World Food Security, means that all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food that meets their food preferences and dietary needs for an active and healthy life. This topic is closely aligned with the Sustainable Development Goals (SDGs), including SDG 1-No Poverty; SDG 2-Zero Hunger; and SDG 3-Good Health and Well-being. Statistica.com showed decrease in Gross National Product of Total production volume of fishing in the Philippines from 2016 to 2021. On the other hand, Philippine Statics Authority census report showed an average on 1.36% growth rate on Philippine population from 2018 to 2021.

FOA.org presented that the total world fisheries and aquaculture production in 2000 (130 million tons), some 74 percent (97 million tons) was available for direct human consumption as food fish. The remainder was reduced into fishmeal and fish oil for use in animal feeding (livestock and aquaculture) or for industrial purposes. Marine capture fisheries production, excluding aquaculture was in the range of 80 to 85 million tons per year in the 1990s. The long-term yearly sustainable yield of marine capture fisheries is estimated at no more than around 100 million tons per year; over-fishing of some species in certain parts of the world's threatens the resource base. Achieving and sustaining these levels assumes more efficient utilization of stocks, healthier ecosystems and better conservation of critical habitats. Inland catches (excluding aquaculture) were recorded at about 7 to 8 million tons per year. However, a recent in-depth study of inland fisheries revealed that actual catches may be double this amount. It is important to note that

fishery resources in many inland water bodies such as rivers and lakes are under increasing environmental threats resulting from continued trends of increasing aquatic pollution, habitat degradation, water abstraction, and other human-made pressures (Food and Agriculture Organization, 2022).

Several municipalities of Camarines Sur, Albay and Catanduanes is located along the coastline Lagonoy Gulf. Fishing and fish value chain is one of the major economic drivers of the community folks. Various type of fish catch is marketed local and even international. This varies from direct human consumption up to processing into various fish-products and other industrial usage. ICT (Information and Communication Technology) cannot contribute to environmental factors in achieving and sustaining efficient enhancement and utilization of fish stocks. Instead, *data analytics* may contribute in planning a sustainable cities and communities along *fish-catch recording and trend visualization* in a certain locality engaged fishing activities. Likewise, data-based decision making for the LGU and other community builders shall benefit in this ***ICT Tool for an Optimized Fishing Repository and Catch Efficiency towards a Sustainable Use and Development.***

Objectives

a. General Objective

Generally, this study aims to design and develop the CATCHFISH - *Collaborative Access to Tracking Community Handling of Fisheries Information, Science, and Harvest*. CATCHFISH is an innovative digital repository that will be able to accurately record information about the fisheries community, fishing resources, and fish catch along the coastline in Lagonoy Gulf.

b. Specific Objectives

Study 1 - Implementation of RDBMS in Establishing a Digital Repository for Community Fisheries Information and Regional Knowledge

1. Gathering and compiling data and information about Fish catch and Fisheries communities.
2. Design the conceptual and logical framework for CATCHFISH Repository.
3. Develop the appropriate tables for the CATCHFISH repository covering all the Fish Catch and Fisheries communities' comprehensive resources and information.

Study 2 - Design and Development of CATCHFISH Front-end Incorporating the Community Participation Modules

1. Design and develop the CATCHFISH as a collaborative tool for fishing resources and catch efficiency integrating functionalities such as a data entry page for content creation, knowledge retrieval features, social network services, and other viewable elements.
2. Evaluation and unit testing of content creation pages, knowledge retrieval pages, social network services pages, and other viewable elements pages.

Study 3 - Empowering Communities through the CATCHFISH Technology Transfer and Training

1. Conduct the usability testing of the system using the ISO 25010:2011 quality model.
2. Conduct capability-building activities with the Fisher folks, communities, and other stakeholders.
3. Formal turnover of the developed CATCHFISH to the stakeholder.

I. METHODOLOGY

Research Design

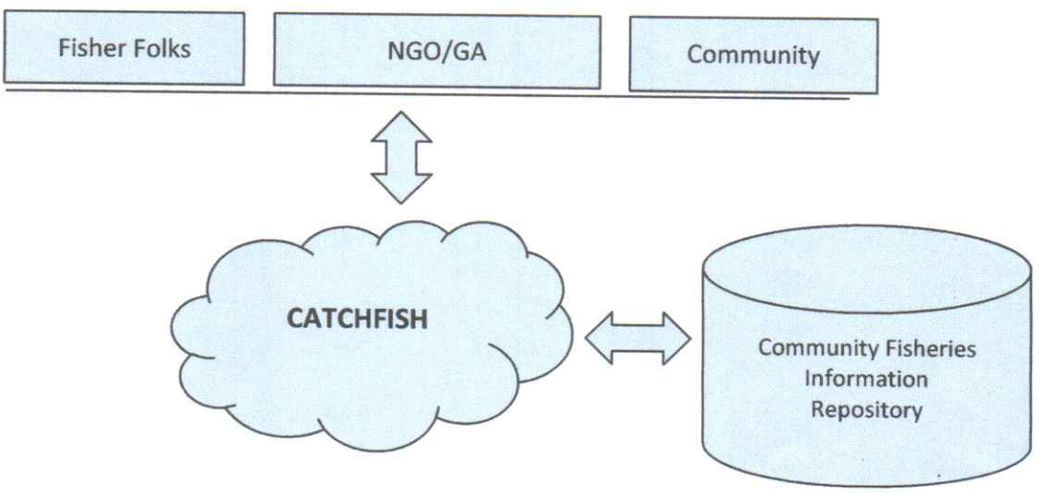
This project utilizes the descriptive and developmental methods of research. It employs the descriptive method because it focuses on the compilation and documentation of the resources about fisheries communities, fishing resources, and fish catch recording data. On the other hand, this project will also employ a developmental method since its primary emphasis is on the development of the CATCHFISH, specifically following the iterative model of software development phases such as requirements analysis, design, coding and implementation, testing, deployment, and operation.

Requirements Analysis

This phase in systems design and development involves data gathering, structured interviews, recording, and identifying data and information the system should accommodate. These include categorizing information related to CATCHFISH and developing functionalities the system will provide. The required tools, technologies, and methods relative to the design and development of CATCHFISH will also evidently be deliberated in this phase.

Design

CATCHFISH will be designed to be a framework for information sharing and management. The repository will be provided with information from various sources.



Coding and Implementation

Identified requirements and functionalities in the requirements analysis phase will be converted into working modules for the information system. Different web frameworks and technologies will be utilized to develop the CATCHFISH. Specifically, researchers will be using PHP, JavaScript and HTML for the frontend and MySQL for the backend,

Testing

This phase is designed to conduct tests that involve potential users of the system where feedback and inputs will be collected to further improve the CATCHFISH. Interviews and surveys will be used. Similarly, developers will conducting an evaluation and unit testing of the content creation pages, knowledge retrieval pages, social network services pages, and other viewable elements pages.

Deployment and Operation

On the deployment of the CATCHFISH, various user types will be implemented to manage the integrity of the information provided by the system. System administrators, content creators, and content moderators will be the entities that will contribute to the repository of data. Regular users will be limited to viewing and accessing the information on the system.

Respondents

Key informants in this study are the fisher folks in Lagonoy Gulf, concerned people from BFAR, researchers and other experts in the field.

II. EXPECTED OUTPUT

A. Product

The primary output of this study is CATCHFISH, a collaborative ICT-based tool for an optimized fishing repository and catch efficiency.

B. Publication

A publishable paper focusing its discussions on the following:

1. Implementation of RDBMS in establishing a digital repository for community fisheries information and regional knowledge
2. Design and Development of Fish Catch Resources Portal with Community Participation Modules

C. Policy

This project will also develop the operational guidelines and procedures for the deployment and proper usage of CATCHFISH to guide users. Similarly, municipal ordinances from the concerned LGUs were also expected.

D. People Services

The researchers will also conduct relevant training with the target users for appropriate usage of the system.

E. Patent/Intellectual Property

A utility model on the CATCHFISH system and copyright of CATCHFISH system source code.

F. Places and Partnerships

This project will foster a strong partnership with the World Wildlife Fund, Local Government Units along Lagonoy Gulf and Partido State University.

III. ESTIMATED BUDGET

PROPOSED WORK AND FINANCIAL PLAN (WFP)					
Line Item, Budget, Particulars, and Specifications	Unit Cost	Quantity	Duration (in months)	PSU Counterpart	WWF Counterpart
Personnel Services (PS)					
Project Lead Honoraria	8,800.00	1	12		105,600.00
Project Members (Study 1&2)	6,000.00	4	6		144,000.00
Research Staff (Study 1&2)	15,000	2	6		180,000.00
Project Members (Study 3)	6,000.00	2	3		36,000.00

Admin Cost (Accounting & Budgeting Staff)	6,000.00		12		72,000.00
Honoraria for Consultants					144,000.00
SUBTOTAL PS				0	681,600.00
Maintenance and Other Operating Expenses (MOOE)					
Transportation Costs	50,000.00	1 lot		50,000.00	
Office Supplies	15,000.00			15,000.00	
Internet Subscription (c/o ParSU Internet Services)	10,000.00	12 months		120,000.00	
SUBTOTAL MOOE				185,000.00	
Equipment Outlay (EO)					
Desktop PCs for Development (Core i7) (c/o IIT Laboratory)	35,000.00	2	Unit	70,000.00	
Upgrading of Hardware:					
Monitor	10,798.00	2	Unit	21,596.00	
UPS	9,598.80	2	Unit	19,197.60	
Kaspersky Internet Security (5 Users)	3,588.00	1	Set	3,588.00	
Wireless Access Point with PoE	8,625.00	1		8,625.00	
8GB RAM 2400MHz	3,000.00	2	Pc	6,000.00	
1TB External HDD Portable	3,598.80	2	Set	7,197.60	
Keyboard	1,500.00	2	Pc	3,000.00	
SUBTOTAL EO				139,204.20	0
TOTAL PS + MOOE + EO:				**324,204.20**	**681,600.00**

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